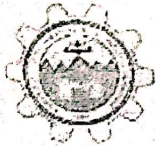


assign out = $S_2? (S_1? (S_0? I_7 : I_6) : I_5) : I_4$;



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Midterm Exam (Spring 2019)

Paper: CSE-308 Digital System Design

Time: 2 Hours

Marks: 25

Note: Attempt all questions on answer sheet.

Question No. 1 (Marks=15) (CLO-1):

- ✓ I. Write a Verilog statement for 8x1 multiplexer using a conditional operator.
- II. Briefly explain how a wand differs from wor.
- ✓ III. Convert the following behavioral code to explicit structural code.

```
module btos(x, a, b);
    input a, b;
    output x;
    wire a, b;
    reg x;
```

always @(a or b) if(a) x = b; else x = ~b;

endmodule

(data flow level)

{ always @(a or b)
if(a)
x = b;
else
x = ~b;

assign x = (a || b) ? b : ~b;

- ✓ IV. How to declare a Verilog constant? Give examples.
- ✓ V. Give the result of each Verilog expression (in binary) for the following inputs: A=4'b0011, B=3'b011 and C=3'b101. Assume A is a 4-bit wire and B and C are each 3-bit wires. Show your results using Verilog notation, such as 3'b101.
- ✓ a. A + (B | C); _____
- ✓ b. ~& A; _____
- ✓ c. (A == B) ? B : C; _____
- ✓ d. {A, {2{C}}; _____
- ✓ VI. Write Verilog statement that declares an 8-bit register, myReg, and initially assigns it the hexadecimal value 29.
- ✓ VII. Write a Verilog statement that declares 8192 x 16-bit memory (myMem).
- ✓ VIII. What is the difference between the design module and the test bench module?
- ✓ IX. Write a Verilog design module for 3 x 8 decoder using behavioural level modelling.
- X. What is the difference between blocking and non-blocking statement? Give examples.
- ✓ XI. Write Verilog code for D flip flop having set and reset ports. The D flip flop should trigger on the negative edge of the clock.
- ✓ XII. What is the difference between positional association and explicit association in specifying ports connectivity? Give examples.
- ✓ XIII. What is the difference between relational operators and bitwise operators in Verilog? Give examples.
- ✓ XIV. Draw the logic circuit diagram that is described by the Verilog code.

```
module M(Q, S, A, B, C);
    output Q;
    input S, A, B, C;
    reg Q;
    always @(posedge C)
        Q <= (S) ? A : B;
endmodule
```

- ✓ XV. What is the difference between initial and always block in Verilog? Give examples.

011
101

111
111

0011
1010

10011
A = 1100
0000

A = 0011
B = 011
C = 101

001 A + (111)

B | C
0 11
1 01

1 11

✓ **Question No. 2 (Marks=5)**

Required

Write the Verilog code for a 4-bit adder/subtractor. An adder/subtractor is a piece of hardware that can give the result of addition or subtraction of the two numbers based on a control signal. Assume that the numbers are in 2's complement notation. Please keep in mind that this is a combinatorial circuit. Also, write a test bench module for the design.

✓ **Question No. 3 (Marks=5) (CLO-3)**

In this module, design a 3-bit gray counter with positive reset. When reset, the count value becomes "000". Recall that a gray counter changes only one bit at a time. For example, a 2-bit gray counter has a count sequence 00, 01, 11, 10 corresponding to decimal count values of 0, 1, 2, and 3 respectively. In the gray count sequence, only one-bit changes between adjacent count values, and the right-most bit is changed as long as it does not result in a code word that has been visited earlier. Also, write a test bench module for the design.

The following are the ports of the module:

CLK	1-bit clock input, all actions must be on the positive edge
RESET	1-bit reset, causes reset on the positive edge
GRAY_OUT	3-bit result

7-11
10, 1, 3, 2, 6, 7, 5, 4, 1